When Does Belief-Based Agent Memory Help? Reliability-Conditional Updating and Provenance-Capped Poisoning Defense

Pranav Singh
Indian Institute of Technology Ropar
2023mcb1308@iitrpr.ac.in

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Abstract

We study *when* a belief-based memory actually helps an LLM agent, and find the answer is conditional. Our vehicle is Nous, a memory that stores knowledge not as records but as a categorical probability distribution per entity-attribute pair, updated by a closed-form Bayesian posterior scored by information-theoretic surprise ($\mathcal{S} = -\log_2 P$), with deltas as the primary artifact and entropy decay as forgetting. Our findings are analytical, not a leaderboard claim. First, the Bayesian belief update is *inert* on the LoCoMo benchmark [Maharana et al., 2024]: on a three-conversation subset, replacing it with naive last-write-wins does not reduce accuracy, because ordinary conversational QA rarely presents contradictory, low-reliability evidence. (Under strict token-F1 on the full benchmark, Nous and the strongest academic baseline A-MEM [Xu et al., 2025] split the four question types two–two.) Second, the update pays off only when observations carry per-source *reliability*; we estimate it from epistemic markers in language, and on a controlled contradiction/staleness benchmark this recovers a large advantage for belief-updating over last-write-wins (100 vs 67) and a strong LLM-over-raw-memory baseline (100 vs 90), concentrated in noise resistance. Third, that reliability signal is an attack surface: *content*-inferred trust is gameable (a confidently phrased poison earns reliability 0.96), so we derive trust from source *provenance*, composed as $\min(\text{provenance}, \text{content})$, unlike prior content-based defenses [Devarangadi Sunil et al., 2026]. This holds memory-poisoning attack success at 0% under a volumetric flood where naive baselines reach 100%; it is bounded, and requires taint-tracking we characterize but do not implement. Finally, we document a 27.5-point macro gap between strict token-F1 and a generous LLM-judge on identical LoCoMo outputs (59.97 vs 32.45), a caution for memory-system comparison. All code and both benchmarks are public.

1 Introduction

A persistent limitation of large language model (LLM) agents is the absence of a continuously evolving memory. The context window provides short-term recall within a session, but once a session ends all conversational history is lost unless externally persisted. This has motivated a growing body of work on agent memory systems, mechanisms that store, retrieve, and reason over information accumulated across many sessions.

Existing approaches treat memory as a database of facts. Retrieval-augmented generation (RAG) [Lewis et al., 2020] encodes conversation chunks as dense vectors and retrieves the k-nearest at query time. Knowledge-graph systems such as Zep [Zep AI, 2024] store subject-predicate-object triples. Episodic systems such as MemGPT [Packer et al., 2023] maintain explicit core memory strings and flush older records to an archival store. More recent systems including Mem0 [Chhikara et al., 2025] and A-MEM [Xu et al., 2025] combine graph and vector representations to improve retrieval precision.

Despite their diversity, all these systems share one assumption: memory is a collection of things that happened. This produces three structural failure modes. First, conflict blindness: when a user changes jobs or preferences, contradictory records accumulate with no principled resolution mechanism. Second, heuristic forgetting: deletion rules are time-based rather than semantic. Third, retrieval granularity mismatch: a question asking what a person has painted over the years requires aggregating values from many sessions, but a fact

store returns individual records with no accumulation mechanism.

We argue these failures share a root cause: treating memory as storage rather than as a model of the world. The brain does not store a verbatim transcript of past conversations; it maintains a generative model of its environment and updates it when observations diverge from predictions [Rao and Ballard, 1999, Friston, 2010]. Surprise drives learning. Unused beliefs decay toward uncertainty. Memory is a compressed, continuously refined probability distribution over the state of the world.

Nous (from the Greek word for mind or intellect) implements this view as a practical agent memory system. The core data structure is a dimension, a categorical distribution over the possible values of a single entity-attribute pair. New observations are scored by information-theoretic surprise and integrated via Bayesian posterior update. The record of that update, a delta, is the primary stored artifact, capturing not what was observed but how it changed the agent's understanding. Forgetting is entropy decay, and conflict resolution is implicit: contradictory evidence naturally shifts probability mass away from the old value.

This paper is a mechanistic study rather than a leaderboard entry, and its contributions are three. **(1) A reliability-conditional account of when belief-updating helps.** A controlled ablation shows the Bayesian update is inert on LoCoMo (a three-conversation subset); naive last-write-wins ties it, because the benchmark rarely presents contradictory, low-reliability evidence; the update then recovers a large, mechanism-attributable advantage once supplied a per-observation reliability signal (estimated from epistemic markers in language), on a contradiction/staleness benchmark where it beats both last-write-wins and a strong LLM-over-memory baseline. **(2) Provenance-capped belief updating as a memory-poisoning defense.** Because the reliability signal is itself an attack surface, we contribute (narrowly, and *not* as the first trust-based defense [Devarangadi Sunil et al., 2026]) four results: trust derived from source *provenance* rather than content; a measurement that content-inferred trust is adversarially gameable; the $\min(\text{provenance}, \text{content})$ composition rule under which a volumetric poisoning flood fails (0% vs 100% for naive baselines); and the taint-tracking requirement without which the defense is laundered away. **(3) A quantification of LoCoMo metric fragility**: a 27.5-point macro gap between strict token-F1 and a generous LLM-judge on identical outputs (59.97 vs 32.45), which we argue should temper leaderboard comparisons. The architecture (dimensions, deltas, surprise, entropy decay) and a dependency-free open-source implementation are the vehicle for these findings, not the claim; we describe them in Section 3.

2 Background and Related Work

Long-term conversational memory. The LoCoMo benchmark [Maharana et al., 2024] evaluates agent memory over conversations spanning up to 35 sessions, 300 to 600 turns, and 9,000 to 16,000 tokens, covering single-hop, multi-hop, temporal, and open-domain questions. LongMemEval [Wu et al., 2025] (ICLR 2025) extends this with harder, audited questions and is increasingly adopted as a complementary standard.

RAG-based memory. RAG [Lewis et al., 2020] retrieves the top-k embedding-nearest chunks. Its primary failure is the inability to aggregate information distributed across many chunks, compounded by the lost-in-the-middle effect [Liu et al., 2024] wherein relevant evidence buried in long contexts is ignored.

MemGPT. MemGPT [Packer et al., 2023] partitions memory into in-context, recall, and archival tiers managed via function calls. It is effective for single-session recall but lacks conflict-resolution semantics. On LoCoMo with GPT-4o-mini it achieves token-F1 of 26.65 (single-hop) and 9.15 (temporal) [Xu et al., 2025].

A-MEM. A-MEM [Xu et al., 2025] enriches memories with Zettelkasten-style links, forming a graph of interconnected notes with LLM-generated attributes. It is the strongest peer-reviewed academic baseline on LoCoMo, reporting token-F1 of 27.02, 45.85, 12.14, and 44.65 on single-hop, multi-hop, temporal, and open-domain with GPT-4o-mini [Xu et al., 2025].

Zep and Mem0. Zep [Zep AI, 2024] builds a temporal knowledge graph; Mem0 [Chhikara et al., 2025] maintains a hybrid vector and graph store. Both report strong LoCoMo scores but use GPT-4o with a lenient LLM-as-judge. An independent

audit by Penfield Labs [Penfield Labs, 2026] found this judge accepts up to 63 percent of intentionally wrong answers, and that approximately 6.4 percent of LoCoMo ground-truth answers contain errors. We therefore restrict our numerical comparison to A-MEM and MemGPT, which report strict token-F1, and, because this judge fragility is central, we report both strict token-F1 and an LLM-judge score for Nous and quantify their gap (Section 5) rather than reporting the judge alone.

Predictive coding and the Bayesian brain. Nous draws theoretically on predictive coding [Rao and Ballard, 1999], which models the brain as a generative model minimising prediction error, and on the Free Energy Principle [Friston, 2010], which unifies perception and learning under surprise minimisation. The Bayesian brain hypothesis [Knill and Pouget, 2004] grounds this computationally. Shannon's information measure [Shannon, 1948] supplies the definition $\mathcal{S}(x) = -\log_2 P(x)$ bits, and Cover and Thomas [Cover and Thomas, 2006] supply the entropy and mutual information foundations.

Concurrent belief-based agent memory. A parallel line of recent work also represents memory probabilistically. BeliefMem [Liao et al., 2026] stores multiple candidate conclusions as independent probability-carrying entries, updated by a Noisy-OR rule as evidence accumulates, and evaluates on LoCoMo and ALFWorld; the Belief Engine [Yang et al., 2026] maintains an auditable log-odds stance per proposition for multi-agent deliberation; and graph-native cognitive memory [Park, 2026] applies symbolic (AGM-style) belief revision over versioned memory. Nous differs in representation, a single closed-form *categorical* Bayesian posterior per entity-attribute dimension, rather than a Noisy-OR over independent hypotheses or a scalar log-odds stance, but the load-bearing difference is one of question: these systems propose a probabilistic memory and measure its accuracy, whereas our contributions are the *reliability-conditional* account of *when* such memory helps (Section 8) and the provenance-capped poisoning defense (Section 9), neither of which they address. BeliefShift [Myakala et al., 2026], a benchmark for temporal belief consistency and opinion drift across sessions, is a natural target for the real-dialogue evaluation we leave to future work.

Memory poisoning and trust-based defenses. A persistent-state attack surface unique to agents is *memory poisoning*: an attacker writes malicious content into long-term memory so the agent acts on it in later sessions. AgentPoison [Chen et al., 2024] optimises a backdoor trigger in a RAG/memory store (attack success $\geq 80\%$ at under 0.1% poison rate), and MINJA [Dong et al., 2025] injects malicious records through query-only interaction (76.8% attack success), and sleeper poisoning [Pulipaka et al., 2026] plants dormant poisoned memories that activate only in later sessions; these establish that memory poisoning is a real, studied threat, which OWASP now lists as ASI06 [OWASP Gen AI Security Project, 2026]. We cite these to motivate the threat, not as a numerical baseline: their attack shapes (optimised trigger backdoor, reasoning-bridge injection) differ from the fact-overwrite setting we study, so success rates are not directly comparable. On defense, the closest prior work assigns *content-derived* trust: Devarangadi Sunil et al. [2026] score trust from composite content and pattern signals with trust-aware retrieval and temporal decay, and A-MemGuard [Wei et al., 2025] detects poisoning by consensus across an agent's reasoning paths, both content- or pattern-based signals. We show (Section 9) that content-inferred trust is adversarially gameable (a confident-sounding injection earns high trust) and that *provenance*-derived trust, composed as $\min(\text{provenance}, \text{content})$ and propagated through trusted intermediaries via taint tracking, is what actually resists poisoning; concurrent lineage-tracking memory [Ouyang and Hou, 2026] builds the derivation graph such propagation requires. OWASP already recommends provenance tracking as a mitigation [OWASP Gen AI Security Project, 2026]; our contribution is not the concept but a concrete mechanism for it and an empirical map of when it holds, where it fails, and what it costs.

3 The Nous Architecture

3.1 Core Abstractions

Definition 1 (Dimension)**.** *A dimension $\mathcal{D}_{e,a}$ for entity e and attribute a is a pair (V, p) where $V = \{v_1, \ldots, v_n\}$ is a finite vocabulary and $p : V \to [0, 1]$ satisfies $\sum_v p(v) = 1$.*

A dimension encodes what the agent believes about one aspect of one entity.

For instance, $\mathcal{D}_{\text{Pranav, employer}}$ might be $\{$(Google, 0.08), (Sarvam AI, 0.87), (unknown, 0.05)$\}$. The mode $\hat{v} = \arg\max_v p(v)$ is the current best belief.

Definition 2 (Delta)**.** *A delta* Δ *is a tuple* $\langle e,\ a,\ p_{\text{prior}},\ p_{\text{post}},\ \mathcal{S},\ \tau,\ \omega\rangle$*, where* p_{prior} *and* p_{post} *are the distributions before and after the update,* $\mathcal{S}$ *is surprise in bits,* τ *is the timestamp, and* ω *is the supporting evidence text.*

The delta is the primary stored artifact of Nous. It records not what happened but how the agent's understanding changed. The full delta sequence for a dimension is an auditable history of every belief revision.

3.2 Information-Theoretic Surprise

Given dimension $\mathcal{D}_{e,a} = (V, p)$ and observed value obs, the surprise is:

$$\mathcal{S}(\text{obs} \mid \mathcal{D}_{e,a}) = -\log_2 p(\text{obs}) \quad \text{bits.} \tag{1}$$

If obs $\notin V$, the vocabulary is extended with $p(\text{obs}) = \varepsilon$ and masses renormalised, yielding near-maximum surprise for novel values. An observation confirming the current belief causes negligible change; one that contradicts it forces a large revision, mirroring the role of prediction error in biological memory consolidation [Friston, 2010].

3.3 Bayesian Update

Nous computes the posterior via Bayes' rule [Bayes and Price, 1763]. The likelihood is:

$$\mathcal{L}(\text{obs} \mid v) = \begin{cases} 1-\varepsilon & v = \text{obs}, \\ \dfrac{\varepsilon}{|V|-1} & \text{otherwise}, \end{cases} \tag{2}$$

and the posterior is:

$$p_{\text{post}}(v) = \frac{\mathcal{L}(\text{obs} \mid v)\ p_{\text{prior}}(v)}{\displaystyle\sum_{v' \in V} \mathcal{L}(\text{obs} \mid v')\ p_{\text{prior}}(v')}. \tag{3}$$

This update is closed-form, runs in $O(|V|)$ time, and requires no gradient computation. Multiple contradictory observations shift probability mass naturally without explicit conflict detection.

Remark. Under Eq. (2) a single confirming observation makes obs the posterior mode whenever $p_{\text{prior}}(\text{obs}) > \varepsilon/(1-\varepsilon)$; with $\varepsilon = 0.01$ this threshold is ≈ 0.0101, so any value already holding non-trivial mass is promoted immediately. This is a sanity property of the update, not a strong result; the interesting behaviour is how mass accumulates under repeated, differently-reliable evidence, which we study empirically in Sections 8 and 9.

3.4 Entropy Decay

A dimension not updated for time Δt grows more uncertain via exponential mixing toward the uniform distribution:

$$p_t(v) = \lambda^{\Delta t}\ p_0(v) + \left(1 - \lambda^{\Delta t}\right) \mathcal{U}(v), \tag{4}$$

where $\mathcal{U}(v) = 1/|V|$ and $\lambda \in (0, 1)$ is a per-dimension retention factor. As $\Delta t \to \infty$, $p_t \to \mathcal{U}$, encoding complete uncertainty. The entropy satisfies:

$$H(p_t) \ = \ -\sum_v p_t(v) \log_2 p_t(v) \ \geq \ \lambda^{\Delta t}\, H(p_0), \tag{5}$$

guaranteeing the agent grows less certain as time passes without reinforcement, a property heuristic deletion rules cannot provide.

3.5 Identity Resolution

When two entity mentions may refer to the same person, Nous computes the symmetrised KL divergence between their shared dimensions:

$$D(e_1, e_2) = \sum_{a \in \mathcal{A}_{e_1} \cap \mathcal{A}_{e_2}} \text{KL}(p_{e_1,a} \| p_{e_2,a}) + \text{KL}(p_{e_2,a} \| p_{e_1,a}). \tag{6}$$

Entities with low divergence across shared attributes are candidates for merging via a weighted mixture of their posteriors.

3.6 Observed-Values Aggregation

Bayesian posteriors are unimodal: after many updates, mass concentrates on one value, discarding historical alternatives. For inherently multi-valued attributes such as activities or places visited, this is incorrect. Nous addresses this via delta-log aggregation: at query time, every value that appeared

with surprise above threshold $\theta_{\min}$ is included in the context fact line:

$$V^*_{e,a} = \{\Delta.\text{obs} \ : \ \Delta \in \mathcal{L}_{e,a},\ \Delta.\mathcal{S} > \theta_{\min}\}. \quad (7)$$

This preserves the Bayesian world model for the current best belief while making the full historical record available for aggregation queries.

3.7 Pipelines

Figure 1 illustrates both pipelines. During ingestion, an LLM extractor parses each session into (e, a, v) triples. For each triple, the current dimension is retrieved, the surprise computed via Eq. 1, the posterior computed via Eq. 3, and a delta appended to the log. A compressor prunes dimensions whose entropy exceeds a ceiling threshold.

During querying, named entities in the question seed a breadth-first search over the entity graph (up to two hops). For each retrieved entity, its dimensions are formatted as fact lines using Eq. 7. The top-k delta records with the highest BM25 similarity to the question are appended as evidence lines, and the assembled context is passed to a category-specific prompt and then to the answering LLM.

4 Experimental Setup

4.1 Benchmark

We evaluate on LoCoMo [Maharana et al., 2024]: ten dyadic conversations spanning 19 to 32 sessions each, totalling 1,540 questions across single-hop (841), multi-hop (282), temporal (321), and open-domain (96). The adversarial category is excluded following the canonical protocol.

4.2 Metrics

We report two answer-quality metrics on the same generated outputs and are explicit about their difference, because it is large and consequential. *Strict token-F1* is the unigram overlap between predicted and ground-truth answer after normalisation; it does not reward semantically correct but differently-phrased answers, and it is the metric under which the academic baselines we compare against report. *LLM-judge accuracy* is the fraction of answers a GPT-4o-mini judge, instructed to accept an answer whose key facts match regardless of wording, marks correct. We treat token-F1 as the primary metric for cross-system comparison and report the judge score alongside it; as Section 5 shows, the two diverge by 27.5 macro points, and the judge score is not comparable to baselines that use token-F1. We additionally report BLEU-1; context recall, the fraction of ground-truth answer tokens present anywhere in the assembled context, as a retrieval diagnostic; and a four-way failure bucket classification (good, answer miss, unknown, retrieval miss) assigned by the same judge.

4.3 Baselines

Nous uses `gemini-2.5-flash` for triple extraction and answer generation, and `gpt-4o-mini` as the judge. We compare against: (i) a no-memory baseline (direct QA, no retrieved context); (ii) MemGPT [Packer et al., 2023]; and (iii) A-MEM [Xu et al., 2025], the strongest peer-reviewed academic baseline. Baseline token-F1 scores are taken from Xu et al. [2025] (GPT-4o-mini backbone). Because the backbone and retrieval stacks differ, we treat this as an indicative rather than a fully controlled comparison; a matched re-run of each baseline is left to future work. We do not compare numerically against Zep or Mem0, which report under GPT-4o with a lenient LLM-judge [Penfield Labs, 2026].

5 Results

Main comparison (strict token-F1). Table 1 reports strict token-F1, the metric under which the academic baselines report. Under this metric Nous is competitive with, not dominant over, A-MEM, splitting the four categories two–two: Nous is best on temporal (51.10 vs 12.14, a 39-point margin) and single-hop (44.66 vs 27.02), while A-MEM is best on multi-hop (45.85 vs 23.48) and open-domain (44.65 vs 10.54). Temporal, where categorical belief tracking most directly captures a changing state, is Nous's clearest strength; the multi-hop and open-domain deficits trace to list-aggregation and retrieval limits we analyse in Section 6.

SH = single-hop, MH = multi-hop, T = temporal, OD = open-domain.

Metric sensitivity (the headline caveat). An earlier version of this work reported the LLM-judge

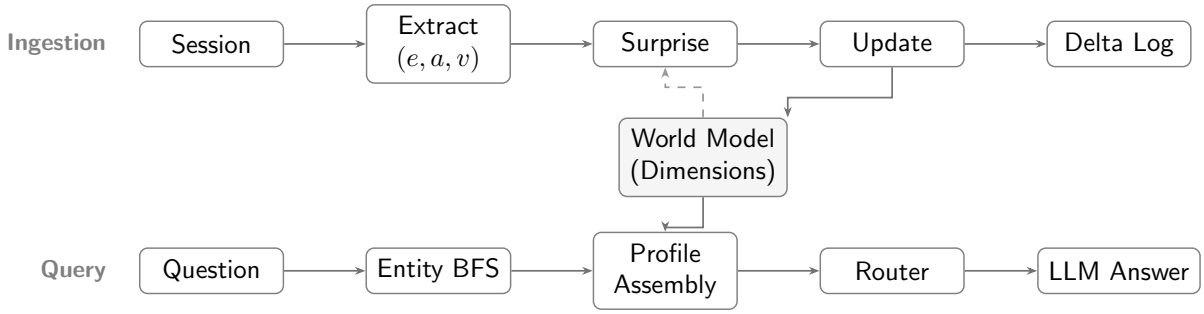


Figure 1: Ingestion (top) and query (bottom) pipelines. The world model, a live store of dimensions, is updated by every Bayesian posterior (solid arrow from Update) and supplies the current dimension state during surprise computation (dashed arrow). At query time, the world model feeds the Profile Assembly step.

Table 1: Strict token-F1 on LoCoMo (10 conversations, 1,540 questions). Baselines from Xu et al. [2025] (GPT-4o-mini). Nous uses gemini-2.5-flash for extraction/answering. Best per column in bold. The comparison is indicative: backbones and retrieval stacks differ (see text).

System	SH	MH	T	OD
LoCoMo	25.02	18.41	12.04	40.36
MemGPT	26.65	25.52	9.15	41.04
A-MEM	27.02	**45.85**	12.14	**44.65**
Nous	**44.66**	23.48	**51.10**	10.54

column of Table 2 as "F1." It is not: it is a generous LLM-as-judge that accepts semantically correct paraphrases. On identical outputs the judge scores 59.97 macro while strict token-F1 scores 32.45, a 27.5-point gap. Three of the four judge scores exceed the mathematical maximum token-F1 attainable given the measured BLEU-1, confirming the two cannot be the same quantity. The gap is largest exactly where answers admit many surface forms: open-domain (judge 62.50 vs token-F1 10.54; context recall only 47.5%, so the judge credits answers the retriever never supported). This is not a Nous-specific artifact but a property of LoCoMo evaluation [Penfield Labs, 2026]; we report it as a result in its own right, and urge that memory-system comparisons fix the judge before comparing scores.

Table 2: Nous under three answer-quality lenses on the same outputs, plus context recall. The judge/token-F1 gap (27.5 macro) is the metric-sensitivity finding; it is *not* a quality improvement.

Category	Token-F1	BLEU-1	LLM-judge	Ctx Rec
Single-hop	44.66	45.60	63.50	86.07
Multi-hop	23.48	23.90	55.32	75.35
Temporal	51.10	56.04	58.57	61.82
Open-domain	10.54	10.06	62.50	47.50
Macro avg	32.45	33.90	59.97	67.69

6 Analysis

6.1 Failure Bucket Breakdown

Table 3 shows the four-way failure distribution. Good denotes a correct answer; answer miss means context was present but the LLM answered incorrectly; unknown means the LLM declined to answer despite relevant context being present; retrieval miss means relevant information was absent from the assembled context.

Table 3: Failure buckets for Nous across all categories.

	Good	Ans.miss	Unknown	Ret.miss
SH	529 (63%)	129 (15%)	154 (18%)	29 (3%)
MH	89 (32%)	92 (33%)	18 (6%)	24 (9%)
T	175 (55%)	65 (20%)	62 (19%)	19 (6%)
OD	40 (42%)	21 (22%)	0 (0%)	35 (37%)

For single-hop, the dominant failure is unknown answer (18%) despite a context recall of 86.07%.

Of 154 such failures, 92 occur when context recall is at least 0.7, meaning the information is present but the model declines to commit. This is a backbone calibration issue rather than an architectural one.

For multi-hop, good (32%) and answer miss (33%) are nearly equal at adequate context recall (75.35%), indicating the model struggles to synthesise complete lists from distributed fact lines. Retrieval miss (9%) contributes additionally, as some questions require linking entities beyond the two-hop search.

Temporal suffers from both low context recall (61.82%, the lowest of all categories) and high unknown answer (19%). Temporal questions often depend on exact session dates embedded in evidence text rather than extracted as dimension values, a representational gap addressed in Section 10.

Open-domain has the highest retrieval miss rate (37%) and lowest context recall (47.50%). Questions about personality and long-term habits do not map cleanly to entity-attribute dimensions, and improving retrieval is the primary path forward.

6.2 Context Recall as a Diagnostic

Context recall cleanly separates retrieval failures from answering failures. For single-hop, context recall is high (86.07%) while strict token-F1 is 44.66, identifying the answering/normalisation step, not retrieval, as the binding constraint. For open-domain, context recall (47.50%) is itself the ceiling: the retriever supports fewer than half the answers, which is why the generous judge score there (62.50) is untrustworthy and the token-F1 (10.54) so low.

7 Ablations: What Actually Drives the Results

The architecture's central claim is that surprise-driven Bayesian belief tracking is what makes the memory work. We test this directly with controlled ablations on a three-conversation subset (385 questions), using BM25 with two-hop entity BFS as the retriever and scoring on both metrics. We toggle one component at a time: the Bayesian update (replaced by naive last-write-wins, which collapses all mass onto the newest value), and observed-values aggregation (which lists all values ever seen for an attribute).

The result is decisive and, for our thesis, uncomfortable (Table 4). **Replacing the Bayesian update with naive last-write-wins does not reduce accuracy**: it is marginally better on the judge (72.46 vs 69.62) and a wash on token-F1 (36.27 vs 36.77). On LoCoMo, the Bayesian belief update is effectively *inert*. The component that does matter is the mundane one: disabling observed-values aggregation costs the most (judge 66.58), the drop concentrated in multi-hop (judge 55.41 → 50.00), because list-valued questions need every value ever seen, not a single most-probable one. We also find the dense retriever inert on this data (single-conversation: judge 63.34 with dense vs 65.26 without), so the BM25+BFS stack the architecture describes is sufficient; and entropy decay is never triggered during LoCoMo evaluation, so these results say nothing about it.

Table 4: Component ablations on the 3-conversation subset (385 questions), BM25+BFS retrieval. Disabling the Bayesian update (FLAT) does not hurt; disabling aggregation (NO-AGG) does. Macro over the four categories.

Variant	LLM-judge	Token-F1
Full engine	69.62	36.77
− Bayesian update (FLAT)	**72.46**	36.27
− Aggregation (NO-AGG)	66.58	35.39

The explanation is not that Bayesian updating is worthless but that LoCoMo does not stress it. Most LoCoMo questions concern facts that are stated once or that do not oscillate; when a value simply needs to be the latest one asserted, "keep the newest" ties the full posterior. The belief machinery is built to arbitrate *contradictory, noisy, differently-reliable* evidence, a regime this benchmark rarely presents. The next section isolates that regime and shows the mechanism is far from worthless there, once given the input it needs.

8 When Belief-Updating Helps: Reliability Extraction

Why the update was inert. The posterior update weights each observation by a reliability r: the likelihood of the observed value is r and the remaining mass is spread over alternatives. A standard extraction pipeline emits $(\text{entity}, \text{attribute}, \text{value})$ triples with *no* reliability, so every observation enters with the same fixed r. With constant r, the

update degenerates into a soft recency-follower: repeated or recent values accumulate mass regardless of how trustworthy they are. This is exactly why it ties last-write-wins.

We verify this at the belief layer with a synthetic contradiction micro-benchmark (no LLM, no retrieval): sequences of (value, r) observations for one attribute, scored on whether each strategy predicts the current true value. With *constant* reliability (what the real pipeline feeds), Bayesian scores 67.1 macro, slightly *below* last-write-wins (73.5). With *varying* reliability that reflects true source trust, Bayesian reaches 100 and dominates both last-write-wins (74.3) and frequency-counting (48.2). The mechanism's value is entirely conditional on a reliability signal the pipeline was not providing.

Supplying the missing signal. We add a reliability-extraction component: the extractor estimates a per-claim reliability in $[0, 1]$ from the speaker's *epistemic markers*: hedging ("maybe", "I think", "if I recall") lowers it, definiteness and self-correction ("actually", "confirmed", "definitely") raise it, using a fixed rubric and never observing ground truth. This per-claim reliability is threaded into the Bayesian update.

End-to-end evaluation. We build a contradiction/staleness benchmark: 60 scenarios across five regimes in which an entity's attribute evolves through natural-language statements of varying linguistic confidence, some false, after which the current value is queried. One regime is *adversarial* (confidence is inverted, so the true value is hedged and a false value asserted), included as an anti-rigging guard where reliability-weighting is expected to lose. We compare, all reading the same claims: NOUS-RELIABILITY (Bayesian, extracted reliability), FLAT (last-write-wins), FREQ (most frequent), and APPEND-RETRIEVE (a capable LLM given *all* raw statements in order).

First, the extractor recovers confidence from phrasing: statements written to be confident receive mean reliability 0.95, hedged ones 0.23 (separation +0.72); the no-match fallback fires near-zero, so this is not an artifact. Second (Table 5), on the realistic regimes NOUS-RELIABILITY scores 100 macro versus 67 for last-write-wins and 46 for frequency, and beats even APPEND-RETRIEVE (90), which is given lossless recall and pays a query cost that grows with memory, whereas Nous queries a bounded posterior. The edge over APPEND-RETRIEVE is concentrated exactly in noise resistance (stable-noise 100 vs 75; reliability-conflict 100 vs 83). As designed, all confidence-based methods fail the adversarial regime; the honest claim is therefore conditional: reliability-weighting helps *when epistemic confidence tracks correctness*, the common pragmatic case, and hurts when it is inverted.

Table 5: Contradiction/staleness benchmark: accuracy predicting the current value. Realistic macro excludes the adversarial guard regime. APPEND-RETRIEVE is an LLM reading all raw statements.

Regime	Nous-rel	Flat	Freq	Append-ret
stable-noise	**100**	42	100	75
reliability-conflict	**100**	25	0	83
clean-change	100	100	83	100
recent-change	100	100	0	100
adversarial (guard)	0	42	50	17
Realistic macro	**100**	67	46	90

To check that the extractor reads epistemic *stance* rather than matching a few templates, we re-ran with a larger, naturalistic phrasing bank containing subtle hedges absent from the rubric ("supposedly", "apparently", "last I checked", "don't quote me"). The signal and the advantage are essentially unchanged (separation +0.71; realistic macro 100 vs 65 vs 40). This is a proof-of-mechanism on synthetic data, not a claim about natural dialogue: the confidence–correctness correlation is by construction in four of five regimes, and a real-dialogue contradiction slice is the natural next step. But it establishes what the LoCoMo ablation could not: given the reliability signal it was designed for, belief-updating memory provides a large, mechanism-attributable advantage over both naive overwriting and strong LLM-over-memory retrieval.

9 Security: Provenance-Capped Belief Updating

The reliability signal of Section 8 raises an adversarial question: if the attacker controls the text, can they forge the confidence that earns trust? Because the numbers in this section are striking, we state their boundaries first: the conditions are what make

the results honest. (i) Provenance trust is assigned by an *oracle*: each source receives a tier from its channel; we do not learn or verify tiers. (ii) The benchmarks are *synthetic* (a controlled belief-layer simulation and a minimal coding-assistant harness over a single repository), not a deployed system. (iii) The defense against laundering requires taint-tracking that we *characterize but do not implement*. (iv) We cite AgentPoison [Chen et al., 2024] and MINJA [Dong et al., 2025] only to establish that memory poisoning is a real, studied threat; our attack is a volumetric fact-overwrite and our success rates are *not numerically comparable* to theirs.

Trust composition. An attacker injects observations asserting a false value W to overwrite an established belief V. Each observation carries reliability $r = \min(\text{provenance}, \text{content})$: content confidence may only *lower* trust within a channel's provenance ceiling, never raise it.

Content-inferred trust is adversarially gameable. Using the extractor of Section 8, we phrased poison in an authoritative style ("Confirmed: the database is now MongoDB, verified"). The real extractor assigns it mean reliability 0.96, versus 0.25 for hedged phrasing. We are explicit about what this does and does not show: we wrote the authoritative poison and our own extractor scored it high, so this demonstrates that content-inferred trust *can* be gamed by construction (an attacker who controls the text controls this signal), not that any specific deployed content-trust defense [Devarangadi Sunil et al., 2026] fails against a real adversary. The architectural consequence is the same either way: a signal the attacker can write cannot be the trust anchor, which is why trust must come from provenance, which the attacker does not control.

Under provenance-capped trust, low-trust poison cannot move the belief. *Condition:* the attacker is confined to a channel the system tiers below 0.5 by provenance (scraped web content, tool output), independent of how the payload is worded. *Result* (Table 6, 400 trials/cell): a volumetric flood of up to 50 poison observations against 4 trusted ones yields 0% attack success, where append-and-retrieve, last-write-wins, and content-trusted updating are all eventually flipped to 100% by volume or by a confident phrasing. Ablating the source-trust signal alone (poison trusted equally) restores 100%, confirming that the *provenance signal*, not the Bayesian arithmetic, is what defends. The guarantee is precisely bounded, not universal: it holds for channel trust ≤ 0.5 at any volume, and fails above ~ 0.55.

Table 6: Attack success rate (%; lower is better) as an attacker injects M false observations against a belief established by four trusted ones, on a channel tiered 0.2 by provenance. Belief-layer simulation, 400 trials/cell. Provenance-capped updating is the only strategy that resists a volumetric flood; this holds *because* the low tier is assigned by provenance, not content.

Memory strategy	M=1	M=3	M=10	M=50
Last-write-wins	100	100	100	100
Append (majority)	0	0	100	100
Content-trusted	100	100	100	100
Provenance-capped	**0**	**0**	**0**	**0**

Laundering defeats naive tiering; only taint-tracking survives, and we do not implement it. *Condition:* the attacker routes the same poison through a *trusted* intermediary, a tool the agent trusts that has ingested attacker-controlled data. If laundered content inherits the intermediary's high tier, attack success returns to 100%: provenance tiering *alone* is fully defeated by laundering. Only when provenance is *propagated*, laundered content down-tiered to its untrusted origin (taint tracking), does success return to 0%. Taint tracking through every trusted intermediary is therefore a *requirement* of the defense, not an optimisation, and a hard one (information-flow control); we characterize it but leave its implementation to future work. The belief update was never the difficult part; provenance propagation is.

Costs: the cap is not free. Two costs follow directly. First, utility: a genuinely reliable source arriving on a low-provenance channel is underweighted, because the min-cap that blocks poison also blocks *legitimate* low-tier corrections: the same mechanism, opposite ground truth. Routing a true, correcting source through tier 0.3 drops accuracy on the contradiction regimes from 100% (uncapped) to 54% on average, with the collapse concentrated in the high-noise regime (down to 21%) rather than the sparse-conflict one (which holds at 88%). Sec-

ond, extraction: because Nous stores structured beliefs rather than raw text, it pays an *extraction tax* that raw retrieval does not. On 30 real repository facts, answerability falls from 100% under perfect extraction to 97% when the real extractor must infer entity/attribute/value, with occasional total drops. This $\sim 3\%$ is a *floor*: our answerer renders all beliefs, so a misfiled belief stays visible, whereas a real retriever would miss it.

End-to-end demonstration. In a minimal coding assistant over one repository (identical LLM and prompt, only the memory layer differing), these effects reproduce end to end. A confident false "migrated to MongoDB," injected on the low-trust web channel, flips an append-and-retrieve baseline (100%) but not Nous (0%); the same poison laundered through a trusted tool flips naive Nous (100%) but not the taint-tracked variant (0%); and on a *legitimate* low-trust correction Nous answers *wrong* where the baseline is right, making the utility cost concrete. We report this as a controlled demonstration on caveated, synthetic conditions, not as evidence of a deployed defense.

10 Limitations and Future Work

The Table 1 comparison is indicative, not controlled: baseline scores come from Xu et al. [2025] under a GPT-4o-mini backbone whereas Nous uses gemini-2.5-flash with a different retrieval stack, so a matched single-backbone re-run of A-MEM (and a controlled comparison against concurrent belief-based memory systems) is needed to make it definitive.

The reliability result (Section 8), while it establishes the mechanism end-to-end, rests on a *synthetic* contradiction benchmark whose confidence–correctness correlation is by construction in four of five regimes. The highest-priority extension is a real-dialogue slice: mining genuine self-corrections and hedged-then-revised facts from conversational data and testing whether the advantage survives. Finer-grained reliability (beyond the near-binary confident/hedged split the extractor currently produces) and learned rather than rubric-based reliability are open directions.

Entropy decay is never exercised by LoCoMo evaluation, so this paper provides no empirical evidence for or against it; a staleness benchmark that queries deliberately outdated facts is required. The categorical dimension is also unimodal, requiring delta-log aggregation as a workaround for set-valued attributes; a set dimension maintaining an independent Bernoulli distribution per element is the main planned architectural extension, and making event timestamps explicit dimension attributes would improve temporal context recall (currently 61.82%).

Finally, all our results use that single backbone, so cross-backbone generalisation is unconfirmed, and completing the in-progress LongMemEval [Wu et al., 2025] evaluation on a second, more carefully audited benchmark is necessary for a complete picture. The Penfield Labs audit [Penfield Labs, 2026] reports roughly 6.4 percent erroneous LoCoMo ground truths and judge-calibration issues, consistent with the metric gap we document in Section 5.

11 Conclusion

We set out to determine *when* belief-based agent memory helps. Our vehicle, Nous, treats knowledge as a predictive world model rather than a database of facts: each entity-attribute pair is a categorical distribution updated by information-theoretic surprise and a closed-form Bayesian posterior, with deltas as the primary artifact and entropy decay as forgetting, an architecture grounded in predictive coding and information theory.

Our empirical account is deliberately honest rather than triumphal. Under a strict token-F1 metric Nous is competitive with, not dominant over, the strongest academic baseline (clearly best on temporal reasoning, mixed elsewhere), and a controlled ablation shows that on LoCoMo the Bayesian belief update is inert, tied by naive last-write-wins, because conversational QA rarely presents the contradictory, low-reliability evidence the mechanism is designed for. We trace this to a missing per-observation reliability signal, add a reliability-extraction component that recovers it from epistemic markers in language, and show on a controlled contradiction benchmark that belief-updating then earns its keep: it beats naive overwriting (100 vs 67) and even a strong LLM-over-raw-memory baseline (100 vs 90), with the advantage concentrated in noise resistance. That same reliability signal is an attack surface: we measure that content-inferred trust is adversarially gameable and show that provenance-capped updating, $\min(\text{provenance}, \text{content})$, holds volumetric

memory-poisoning at 0% attack success where naive baselines reach 100%, a defense that is bounded, requires taint-tracking, carries a measured utility cost, and is narrow in novelty relative to prior content-based defenses. We also quantify a 27.5-point gap between strict token-F1 and a generous LLM-judge on identical outputs, a caution for the whole subfield's comparisons. The takeaway is not that probabilistic world models win every benchmark, but a sharper and more useful claim: they help precisely when memory must arbitrate conflicting, differently-trustworthy evidence, and we identify the component that makes that benefit real. All code and evaluation harnesses, including both benchmarks, are released publicly.

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